

Case Study Blueprint: Global Food Price Data Ingestion & Engineering Pipeline

1. Project Background & Objective

The organization requires a clean, structured, and reliable data repository derived from raw global food pricing files (`Food_Prices.csv`). The current raw data contains unstandardized structural formats, unformatted currency attributes, potential duplicate records, and extreme statistical outliers that distort analytical models.

The objective of this project is to architect an end-to-end data engineering pipeline that ingests the raw data, applies strict data hygiene standards, segments historical information, and automates executive-ready reporting deliverables.

2. The Step-by-Step Project Roadmap

Step 1: Environment Setup & Raw Data Ingestion

- **Objective:** Establish a secure, permanent workspace and ingest the raw source file.
- **Action Items:** Allocate a dedicated, permanent database library to store all project assets.
 - Programmatically import the raw CSV dataset, ensuring that the system scans the entire file structure to accurately determine appropriate variable types and lengths without truncating data.

Step 2: Schema Standardization & Variable Normalization

- **Objective:** Conform the dataset to standardized database naming conventions and business rules.
- **Action Items:** Identify and rename columns containing spaces, special characters, or illegal syntax into clean, standard database formats.
 - Implement a system to map numeric calendar representations (1–12) into human-readable text month designations.
 - Apply corporate financial currency formatting to all USD and native price fields to ensure executive readability.

Step 3: Data Quality Assurance & Anomaly Cleaning

- **Objective:** Audit the data for completeness and isolate anomalies that could skew downstream analytics.
- **Action Items:** Profile the dataset to locate and log missing records across key variables.
 - Identify and remove exact duplicate records based on a logical composite key (e.g., combination of geography, timeline, and product).
 - Establish a statistical framework (such as the Interquartile Range method) to programmatically calculate upper and lower bounds for pricing fields. Isolate and remove severe outliers into separate data views.

Step 4: Data Mart Segmentation

- **Objective:** Categorize and slice the data to optimize performance for regional and historical analysis.
- **Action Items:** Partition the master dataset into independent, historical subsets segregated by individual years.
 - Extract distinct, sorted indices of unique core dimensions (such as unique countries and active historical years) to serve as lookup references.

Step 5: Automated Executive Reporting Engine

- **Objective:** Deliver the finalized data assets in a dynamic, stakeholder-friendly format.
- **Action Items:** Construct a multi-tab spreadsheet generation pipeline that automatically outputs a single workbook file.
 - Configure the export layout to cleanly separate the master cleaned data, the unique country index, and the timeline index into individual named tabs.
 - Ensure quality-of-life report features are enabled, such as embedded titles, frozen column headers, and visible column labels.

3. Core Deliverables Required for Sign-Off

To successfully close this project, the pipeline must successfully output:

1. **A Clean Master Dataset:** Free from naming defects, properly formatted, with outliers taken care of,
2. **An Executive Workbook:** A dynamic, multi-tab spreadsheet with locked headers, clean formatting, and structured naming.